MOEZ BOUASSIDA

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WORK EXPERIENCE

UIB – Société Générale Group Tunis, Tunisia

Data Scientist (AI Engineering) | May 2025 – Present

- Designed and deployed on-premise GenAI systems to production: a RAG assistant over Central Bank of Tunisia regulatory documents (self-hosted Qwen LLM, Qdrant vector store, MCP) and an agentic text-to-SQL system (LangGraph) turning business data requests into SQL for data engineering teams
- Automated Basel III and CRR3 regulatory reporting pipelines for ECB and BCT submissions, cutting finance and risk teams' preparation time from a full week to under 1 hour (CRR3 with Société Générale's French team)
- Contributing to the bank's AI architecture: model serving on local infrastructure, data governance and the technical foundations to scale AI adoption across teams
- Built production ETL workflows in Alteryx and Oracle SQL with engineers and domain experts, resolving data quality issues in banking data reporting and extraction

Faits Mains Remote

AI Engineer (Freelance) | Apr 2025 – Jun 2025

- Built a multimodal RAG chatbot with Claude and Pinecone, deployed serverless on AWS (Lambda, API Gateway, S3, DynamoDB, CloudWatch) with caching, rate limiting and CI/CD (GitHub Actions) to keep running costs low
- Designed a serverless image generation API with Gemini, including prompt safety filtering and cloud-native observability

OPTOSCOPIA Tunis, Tunisia

Computer Vision Engineer | Sep 2024 – May 2025

- Led the development of an object detection model for sperm vitality assessment (91% accuracy) and a segmentation model for sperm morphology on microscopy images of semen samples, deployed in production across 5+ clinical sites
- Built preprocessing pipelines for large-scale microscopy image data (normalization, augmentation, dataset curation)
- Optimized inference with ONNX and TensorRT for real-time analysis on clinical hardware
- Defined annotation protocols and trained biologists on expert labelling, improving dataset quality and annotation consistency by 30%

WSK Medical Amsterdam, Netherlands

AI Research Intern – Digital Pathology | Mar 2024 – Aug 2024

- Researched deep learning approaches for automated Tumor-Stroma Ratio (TSR) scoring on H&E-stained whole-slide images (WSI) of colorectal cancer (ARABESC project), including a literature review of TSR methods and Vision Transformer encoders for computational pathology
- Trained and benchmarked semi-supervised learning models to learn from limited pathologist annotations, comparing several semi-supervised techniques across CNN, ViT and hybrid CNN-ViT architectures
- Explored domain transfer of a self-supervised (SSL) foundation model pretrained on prostate histopathology to colorectal cancer, producing tissue heatmaps over our whole-slide images
- Developed a desktop application (Electron, OpenSlide) for gigapixel WSI visualization with region-of-interest selection and on-demand AI inference

EDUCATION

National School of Computer Science (ENSI), Tunisia

Engineering Degree in Computer Science (Master's level) | Sep 2021 – Sep 2024

Preparatory Institute for Engineering Studies of Tunis (IPEIT), Tunisia

Preparatory Cycle for Engineering Schools, Mathematics and Physics | Sep 2019 – Jun 2021

PROJECTS

SwinUNETR 3D Brain Tumor Segmentation — PyTorch, MONAI, LangGraph, WebGL, AWS, Terraform
github.com/moebouassida/SwinUNETR-3D-Brain-Segmentation
Trained SwinUNETR v2 on BraTS 2021 multi-modal MRI: 0.8961 mean Dice on 188 held-out test cases (HD95 5.62 mm; the MONAI paper reports ~0.875 on its validation set). Built gradient saliency and MC Dropout uncertainty maps, a LangGraph multi-agent system for clinical reporting, a WebGL 3D viewer, federated learning (Flower) and AWS deployment via Terraform, documented as a high-risk system under the EU AI Act with human oversight.

Multi-Agent Radiology Report Generation — LangGraph, Claude Sonnet, Qdrant, FastAPI, PostgreSQL, AWS ECS Fargate, Terraform
github.com/moebouassida/Multiagent-Radiology-Report
Four LangGraph agents (image analysis, RAG clinical context, report drafting, QA validation with automatic re-drafting) turn DICOM chest X-rays into structured reports, reaching a 0.91 QA validation score (completeness, urgency, consistency and hallucination checks). Includes Grad-CAM heatmaps (TorchXRayVision), an MCP server for prior patient reports, mandatory radiologist review before any report is final (EU AI Act Art. 14), and GDPR controls (PII stripping, anonymization, audit trail, right to erasure); deployed on AWS ECS Fargate via Terraform.

OmicPath: Multimodal Breast Cancer Survival Prediction — PyTorch, UNI2-h, OpenSlide, FastAPI, Docker, Prometheus
github.com/moebouassida/OmicPath
Fused gigapixel H&E whole-slide images (256×256 tiling, UNI2-h pathology foundation model features, attention-based multiple instance learning) with RNA-seq pathway scores via bidirectional cross-attention to predict survival, PAM50 subtype and TP53 status. On TCGA-BRCA (1,030 patients, 5-fold CV): C-index 0.666 ± 0.053, PAM50 accuracy 0.70, TP53 AUROC 0.818, with ABMIL attention heatmaps for spatial interpretability.

Path-VQA MedGemma Fine-Tuning — Hugging Face, PEFT, W&B, FastAPI, Gradio
github.com/moebouassida/Path-VQA-Med-GaMMa-Fine-Tuning
Fine-tuned MedGemma 4B (SigLIP vision encoder) with DoRA + RSLoRA (0.9% trainable parameters, bfloat16) on PathVQA Enhanced (19.6K training pairs of a 32.6K-pair dataset), raising yes/no accuracy from ~58% to ~72% (estimated from the training run) on an H100; adapter and dataset published on Hugging Face.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Java, JavaScript
Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, timm, MONAI, ONNX, TensorRT
Digital Pathology: whole-slide image analysis (OpenSlide, QuPath), H&E histopathology, pathology foundation models (UNI2-h), multiple instance learning (ABMIL), semi- and self-supervised learning, Vision Transformers, microscopy image analysis
Computer Vision: OpenCV, object detection, image segmentation, OCR, vision-language models (MedGemma, Qwen-VL)
Generative AI: LLM APIs (Claude, OpenAI, Gemini), open-source LLMs (Qwen), RAG (Qdrant, Pinecone), LangGraph and LangChain agents, MCP, fine-tuning (LoRA/DoRA, Unsloth), Ollama
Data: NumPy, Pandas, SciPy, scikit-learn, Oracle SQL, Alteryx
Deployment: AWS (EC2, ECS Fargate, Lambda, S3, DynamoDB, CloudWatch), Terraform, Docker, CI/CD (GitHub Actions), FastAPI, Electron, Prometheus, W&B, MLflow, Git

CERTIFICATIONS, LANGUAGES & AWARDS

Languages: Arabic (native), French (bilingual), English (C1), German (A2)
Certifications: Project Management; Agile and Scrum
Award: 1st place, ENSI PCD Challenge 2023 (4th edition), for an image-based fashion recommendation system using deep visual similarity and personalized search